



# LIDIA SCUDIERO

DATA SCIENCE | NEUROSCIENCE | MACHINE  
LEARNING | DECISION SUPPORT | EXPLAINABLE AI

## PROFILE

Big Data and AI graduated, with experience in data science and applied research. I build analytics pipelines from data cleaning and feature engineering to dashboards and report. I am particularly interested in using data, AI and behavioural insights to support evidence-based decisions in complex technical environments.

## CONTACT INFORMATION

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**Github:** [lidia\\_scudiero\\_github](#)

**Portfolio:** [lidia\\_scudiero\\_portfolio](#)

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## SELECTED IMPACT

- **Challenged an established assumption with real-world data** analysing **7.5M IoT telemetry records** and **cross-referenced** them with technical documentation, **finding that 99% of recorded failures appeared as sudden events**, prompting a review of whether predictive modeling was appropriate.
- **Connected fragmented sources** linking attack datasets, Reddit discussions, scientific literature, and EU funding information using knowledge graphs, RAG, time-series, and geospatial analysis to **investigate relationships between technology, public perception, and research priorities**.
- **Recognised for academic excellence** selected by the **Italy-USA Foundation** among outstanding Italian graduates in Cognitive Neuroscience; preselected for the **ESA Graduate Trainee Programme** (Data Quality), 2026.

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## PROFESSIONAL EXPERIENCE

### LLM EVALUATION AND QUALITY SPECIALIST | DataAnnotation Tech | Remote | May 2026 - Present

- **Evaluate** LLM outputs against defined quality and **reasoning criteria**, **identifying biases**, inconsistencies, and **instruction-following failures**.
- Apply **structured evaluation** and **prompt-engineering** principles to support systematic **quality control** and **evidence-based** assessment of AI outputs.

### DATA SCIENCE INTERN | Almawave S.p.A. | Remote | Sep 2025 - Dec 2025

- Conducted a **feasibility study** for **predictive maintenance of IoT-enabled ticket gates**, analysing **7.5M telemetry records** to assess failure patterns, data quality, and model applicability.
- **Found** that approximately **99% of recorded failures appeared as sudden events**. Cross-referenced telemetry with **technical documentation** and investigated whether the discrepancy reflected the underlying system, available signals, or how failures were defined and recorded.
- **Built an end-to-end analytical pipeline** for data cleaning, KPI computation, anomaly/outlier detection, and exploratory analysis with **machine learning and deep learning models**.
- Translated findings and modelling limitations into **interactive dashboards and reports** for technical and non-technical stakeholders, and proposed alternative monitoring and data-governance strategies.

## CIVIL SERVICE VOLUNTEER| Social Services Office, Giffoni Sei Casali| June 2023 - June 2024

- Provided **digital assistance and front/back office** support, collecting and managing **user data for social projects**.
- Supported vulnerable people and **contributed to basic impact analysis activities**, strengthening teamwork, communication and problem-solving skills.

## WEB EDITOR AND PRIVATE TUTOR| Salerno, Mondadori S.p.A-formerly Banzai media o2.o.it| 2015-2024

- **Researched** and wrote online articles on **technology, science, and education**, translating complex topics into clear and **accessible** content for non-expert audiences.
- Developed strong **research, synthesis, and communication skills** through **independent topic research** and long-term **writing experience**.
- Delivered private lessons to primary and secondary students, **strengthening communication and audience-adaptation skills**.

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## SELECTED PROJECTS (more on my portfolio and github)

### Motor Imagery BCI - EEG Decoding

- Developed an **end-to-end EEG decoding pipeline** for **motor-imagery BCI**, covering preprocessing, spatial filtering, and machine/deep-learning classification.
- Evaluated **signal quality, interpretability, and subject-independent generalisation**, investigating limitations relevant to **robust BCI applications**.

### Alzheimer's Clinical Support System

- Developed an **interpretable clinical decision-support system** for dementia-risk stratification using heterogeneous clinical and lifestyle data.
- Investigated class imbalance and demographic bias, compared tree-based models, and used SHAP to support **evidence-based interpretation of model predictions**.

### Cybersecurity Analytics and Multi-Source Research

- Combined attack datasets, Reddit discussions, scientific literature, and **EU funding information to investigate relationships between cyberattacks, public perception, and research priorities**.
- Used knowledge graphs, RAG, entity extraction, time-series, and geospatial analysis to **connect fragmented information and identify patterns across heterogeneous sources**.

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## TECHNICAL SKILLS

**Programming:** Python, SQL

**Data analysis and BI:** Advanced Excel, Power BI, R, SPSS, Visual Studio, Streamlit, DuckDB

**Data science Tools :** Pandas, Numpy, Scipy, Matplotlib, Seaborn, Altair, Folium, Plotly, scikit-mobility, Shapely, OSMnx, networkx

**ML and AI :** Scikit-Learn, Keras, TensorFlow, sktime, NLTK, Spacy(NLP)

**Neuroscience and Signal Processing:** MNE-Python, PyLSL

**Web and Data Crawling:** Selenium, BeautifulSoup

**Generative AI:** Hugging Face, Mistral, Cohere

**Collaboration and other tools:** Notion, Docker, Git, Canva, Inkscape, Psytoolkit, Social media

## LANGUAGES

Italian - Native, English -Advanced

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## EDUCATION

**Second-level Master in Big Data Analytics and Artificial Intelligence for Society | University of Pisa| 2024-2026**

- Final Grade: Excellent
- Data mining, machine learning, data visualization, data journalism, web scraping, NLP, information retrieval, social network analysis.

**MSc in Cognitive Psychology | University of Campania "Luigi Vanvitelli" | 2020-2023**

- Curriculum : Cognitive Neuroscience
- Final Grade: 110 cum Laude

**BSc in Psychological Sciences and Techniques | University of Campania "Luigi Vanvitelli" | 2017-2020**

- Final Grade: 110 cum Laude

**Merit-based Scholarship (2017-2021)** for academic excellence and high academic performance during university studies.